
SALMODEX SYRUP

Composition and Strength of Active Ingredients:

Each 5 ml contains:

Salbutamol	2 mg
(2.41mg of Salbutamol Sulphate equivalent to 2mg Salbutamol)	
Bromhexine Hydrochloride	4 mg
Preservative:	
Sodium Benzoate	0.2% w/v

Product Description

Light pink liquid with cherry flavour.

Pharmacodynamics

Salbutamol is a synthetic sympathomimetic amine. Salbutamol stimulates beta-adrenergic receptors and has little or no effect on alpha adrenergic receptors. Beta-adrenergic agonists stimulate the production of cyclic adenosine-3',5'-monophosphate(AMP) by activation of the enzyme adenylyl cyclase.

Cyclic AMP appears to have greater stimulating effect on beta receptors of the bronchial, uterine and vascular smooth muscles (beta-1 receptors).

Bromhexine is a mucolytic agent used in the treatment of respiratory disorders associated with viscid or excessive mucus.

Pharmacokinetics

Salbutamol sulfate is rapidly and well absorbed following oral administration.

Peak plasma salbutamol concentrations occur within 2.5 and 2 hours following administration of the conventional tablets and oral solution respectively. Following oral administration of 2 mg of salbutamol every 6 hours as conventional tablets in healthy individuals, steady-state peak plasma salbutamol concentrations average 5.3-6.8 and steady-state trough plasma concentrations minutes 3.8-4.3 ng/mL. Bronchodilation begins within 30 minutes after oral administration of conventional tablets, with peak effect in 2-3 hours, and may persist up to 4-6 hours.

Results of animal studies indicate that salbutamol does not cross the blood-brain barrier, but the drug apparently crosses the placenta. After oral administration, the plasma half-life is reportedly 2.7-5 hours.

Following oral administration of salbutamol sulfate to healthy individuals, about 75% of a single dose is excreted in urine within 72 hours, mainly as the major metabolite; about 4% of the dose is excreted in feces.

Bromhexine hydrochloride is rapidly absorbed from the gastro-intestinal tract and about 85 to 90% of a dose is excreted in the urine mainly as metabolites of bromhexine. Bromhexine is highly bound to plasma proteins.

Indications

A bronchodilator with mucolytic expectorant for clearance of viscid purulent sputum in acute and chronic bronchiectasis and whooping cough.

Recommended Dose

Adults: 5ml, 3 to 4 times a day

Children below 6 years: 1.25 to 2.5 ml, 3 to 4 times a day.

Older children: 2.5 to 5ml, 3 to 4 times a day.

Or as directed by the physician.

Route of Administration

Oral.

Contraindications

Thyrotoxicosis, angina pectoris, severe cardiovascular diseases and peptic ulcers.

Warning and Precautions

Patients should be warned that if either the usual relief is diminished, or the usual duration of action reduced, they should not increase the dose or its frequency of administration, but should seek medical advice.

Salbutamol oral preparations and non-selective beta-blocking drugs such as Propranolol, should not usually be prescribed together.

Salbutamol should be administered cautiously to patients suffering from thyrotoxicosis.

Potentially serious hypokalemia may result from beta-2 agonist therapy.

Particular caution is advised in acute severe asthma as this effect may be potentiated by concomitant treatment with xanthine derivatives, steroids, diuretics and by hypoxia.

Tocolysis: Serious adverse reactions including death have been reported after administration of terbutaline/ salbutamol to women in labor. In the mother, these include increased heart rate, transient hyperglycaemia, hypokalaemia, cardiac arrhythmias, pulmonary oedema and myocardial ischaemia. Increased fetal heart rate and neonatal hypoglycaemia may occur as a result of maternal administration.

Since mucolytics may disrupt the gastric mucosal barrier, bromhexine should be used with care in patients history of peptic ulceration.

Very rare cases of chronically associated severe skin impairments such as Stevens Johnson Syndrome, Toxic Epidermal Necrolysis (TEN), Erythema Multiforme (EM) and Acute Generalized Exanthematous Pustulosis (AGEP) have been reported. In most cases, these could be explained by the severity of the underlying disease or concomitant administration of another drug. In the early stages of such severe skin reactions, initially only nonspecific flu-like symptoms appear, e.g fever, arthralgia, runny nose, cough, and sore throat. If skin or mucous membrane damage occurs, seek medical advice immediately and discontinue treatment as a precaution.

Interactions with Other Medicaments

Beta-blockers inhibit the bronchodilator effect of sympathomimetic bronchodilators. Diuretics and xanthines may augment hypokalemia. Hypokalemia associated with high doses of salbutamol may result in increased susceptibility to digitalis induced cardiac arrhythmia.

Pregnancy & Lactation

Administration of salbutamol during pregnancy should only be considered if the expected benefit to mother is greater than any possible risk to fetus.

As salbutamol is probably secreted in breast milk, its use in nursing mothers is not recommended unless the expected benefits outweigh any potential risk.

Side Effects

Tachycardia, palpitation, precordial pain, nervousness, tremors in a few patients. Upper gastrointestinal complaints like nausea, vomiting may be observed in a few cases.

Immune System Disorders

Frequency not known: Anaphylactic reactions including anaphylactic shock.

Skin and Subcutaneous Skin Disorders

Frequency not known: Severe skin reactions (including Stevens Johnson syndrome, Toxic epidermal necrolysis (TEN), Erythema Multiforme (EM) and Acute Generalized Exanthematous Pustulosis (AGEP)).

Symptoms and Treatment of Overdose

Salbutamol: The preferred antidote for overdosage with salbutamol is a cardioselective beta-blocking agent, but beta-blocking drugs should be used with caution in patients with a history of bronchospasm. Hypokalemia may occur following overdosage with salbutamol. Serum potassium levels should be monitored.

Effects on Ability to Drive and Use Machine

No studies have been performed with Salbutamol and Bromhexine on the effect of the ability to drive and use machines. However, this product is not likely to cause drowsiness or to make you feel dizzy either. However, if you experience, stop consuming the product and seek for immediate medical attention.

Instructions for Use

Always consume exactly as advised by your physician. You should check with your doctor or pharmacist if you are unsure.

Taking this medicine:

This product is only to be taken by mouth.

Shake the bottle well before measuring the dose.

To make sure that you use the measuring cup provided to measure your dose

If you prefer, you can ask your pharmacist for a 5 ml plastic spoon or an oral syringe to use instead

Remove the measuring cup from the bottle cap before use and fill up according to the measuring dosage line.

Rinse the measuring cup with water after use.

This product can be taken with or without food

How much to take

Adults: 5ml, 3 to 4 times a day

Children below 6 years: 1.25 to 2.5 ml, 3 to 4 times a day.

Older children: 2.5 to 5ml, 3 to 4 times a day.

Drink water after completing the measured dose.

Storage Condition

Keep the bottle tightly closed protected from light. Store below 30°C.

Presentation:

1. 60ml light pink liquid with cherry flavour filled in 60ml amber PET bottle with white HDPE screw cap. A measuring cup is put on top of the capped bottle. The filled bottle is packed in a pre-printed carton.
2. 100ml light pink liquid with cherry flavour filled in 100ml amber PET bottle with white HDPE screw cap. A measuring cup is put on top of the capped bottle. The filled bottle is packed in a pre-printed carton.

Registration Number:

MAL

Manufactured by & Product Registration Holder:

KCK Pharmaceutical Industries Sdn. Bhd.

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